The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section D. Eigenvalue equations and observables

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Outline

- A. One-particle wave function space
- B. State space and Dirac notation
- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $|\boldsymbol{r}\rangle$ and $|\boldsymbol{p}\rangle$; operators $\boldsymbol{R}$, $\boldsymbol{P}$
- F. Tensor products and applications

Eigenvalues and eigenvectors of an operator

Definition

A ket $|\psi\rangle$ is said to be an eigenvector (or eigenket) of the linear operator A if there exists a (possibly complex) number λ such that:

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If $A|\psi\rangle = \lambda|\psi\rangle$, then for any complex number α:

$$A(\alpha|\psi\rangle) = \alpha A|\psi\rangle = \alpha\lambda|\psi\rangle = \lambda(\alpha|\psi\rangle),$$

so any scalar multiple of $|\psi\rangle$ is also an eigenvector of A with the same eigenvalue.

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$$A|\psi_i\rangle = \lambda|\psi_i\rangle, \qquad i = 1, 2, \ldots, g.$$

Any linear combination $|\psi\rangle = \sum_{i=1}^{g} c_i|\psi_i\rangle$ is also an eigenvector of A with the same eigenvalue λ.

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Inserting the closure relation $\sum_j |u_j\rangle\langle u_j| = \mathbb{1}$:

$$\sum_j \langle u_i|A|u_j\rangle\langle u_j|\psi\rangle = \lambda\langle u_i|\psi\rangle.$$

Finding the eigenvalues and eigenvectors of an operator

$$\sum_j \langle u_i|A|u_j\rangle\langle u_j|\psi\rangle = \lambda\langle u_i|\psi\rangle.$$

Defining

$$A_{ij} = \langle u_i|A|u_j\rangle, \qquad c_j = \langle u_j|\psi\rangle,$$

we obtain the system:

$$\sum_j A_{ij}c_j = \lambda c_i \quad \text{or} \quad \sum_j (A_{ij} - \lambda\delta_{ij})c_j = 0.$$

The characteristic equation

The homogeneous system

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has non-trivial solutions only if the determinant of the coefficients vanishes:

$$\boxed{\det(A - \lambda I) = 0.}$$

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This is the **characteristic (or secular) equation** of A. It is an Nth-order polynomial in λ, so it has N (possibly repeated) roots, real or complex:

$$\lambda_1,\ \lambda_2, \ldots, \lambda_N.$$

The characteristic equation

Important:

The characteristic equation is independent of the basis chosen. Therefore, the **eigenvalues of an operator are the roots of its characteristic equation.**

Simple and multiple eigenvalues

Let λ_0 be a root of $\det(A - \lambda I) = 0$. To find the eigenvectors associated with λ_0, substitute it into

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If λ_0 is a simple root, the system consists of $(N - 1)$ independent equations — there is only **one independent eigenvector** (non-degenerate case).
If λ_0 is a multiple root of order p, the system contains $(N - p)$ independent equations. Then λ_0 is p-fold degenerate, and the corresponding eigenspace has dimension p:

$$\dim E_{\lambda_0} = p.$$

Construction of eigenvectors

The general eigenvector corresponding to λ_0 can be written as:

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For a simple root λ_0, all coefficients c_i are proportional to a single one; we can write

$$|\psi_{\lambda_0}\rangle = c_1 |v_{\lambda_0}\rangle,$$

with normalized $|v_{\lambda_0}\rangle$.

Construction of eigenvectors

For a p-fold degenerate eigenvalue λ_0, there exist p independent solutions $\{|\psi_{\lambda_0}^{(r)}\rangle\}$. Any linear combination

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Hence, all vectors with the same eigenvalue form a p-dimensional eigenspace E_{λ_0}.

Hermitian operators and degeneracy

If A is **Hermitian**, all its eigenvalues are real. Moreover, eigenvectors belonging to different eigenvalues are orthogonal:

$$\langle \psi_i | \psi_j \rangle = 0 \quad \text{if } \lambda_i \neq \lambda_j.$$

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For a p-fold degenerate eigenvalue λ_0, the corresponding eigenvectors $\{|\psi_{\lambda_0}^{(r)}\rangle\}$ span a p-dimensional vector subspace E_{λ_0}. They can always be chosen orthonormal:

$$\langle \psi_{\lambda_0}^{(r)} | \psi_{\lambda_0}^{(s)} \rangle = \delta_{rs}.$$

Hermitian operators and degeneracy

Summary:

- λ_0 simple root $\Rightarrow$ one eigenvector (nondegenerate).
- λ_0 multiple root of order $p \Rightarrow p$ independent eigenvectors.
- For Hermitian A: eigenvectors with distinct eigenvalues are orthogonal, and degenerate ones can be orthonormalized.

Properties of Hermitian operators

A Hermitian operator satisfies:

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(i) Eigenvalues are real.

Taking the scalar product of $A|\psi\rangle = \lambda|\psi\rangle$ with $\langle\psi|$:

$$\langle\psi|A|\psi\rangle = \lambda\langle\psi|\psi\rangle.$$

Since $\langle\psi|A|\psi\rangle$ is real for Hermitian A, we obtain:

$$\langle\psi|\,A\,|\psi\rangle^* = \langle\psi|\,A^\dagger\,|\psi\rangle = \langle\psi|\,A\,|\psi\rangle \qquad \boxed{\lambda = \lambda^*.}$$

Hence, all eigenvalues of a Hermitian operator are real.

Properties of Hermitian operators

(ii) Eigenvectors with different eigenvalues are orthogonal.

For $A|\psi\rangle = \lambda|\psi\rangle$ and $A|\varphi\rangle = \mu|\varphi\rangle$, we have:

$$\langle\varphi|A|\psi\rangle = \lambda\langle\varphi|\psi\rangle, \qquad \langle\varphi|A|\psi\rangle = \mu\langle\varphi|\psi\rangle.$$

Subtracting:

$$(\lambda - \mu)\langle\varphi|\psi\rangle = 0 \Rightarrow \boxed{\langle\varphi|\psi\rangle = 0 \text{ if } \lambda \neq \mu.}$$

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Eigenvectors from different eigenspaces are orthogonal:

$$\langle\psi_n^i|\psi_{n'}^j\rangle = 0 \quad \text{if } n \neq n'.$$

Inside each eigenspace they can be chosen orthonormal:

$$\langle\psi_n^i|\psi_{n'}^{i'}\rangle = \delta_{nn'}\delta_{ii'}.$$

Definition of an observable

Definition

A Hermitian operator A is an observable if this orthonormal set of eigenvectors $\{|\psi_i^{(r)}\rangle\}$ forms a basis of the state space.

Orthonormal system and closure relation

If the set $\{|\psi_n^i\rangle\}$ forms a complete basis, then the **closure relation** holds:

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Eigenvalues a_n with degeneracy g_n correspond to eigenspaces $\mathcal{E}_n$ of dimension g_n spanned by $\{|\psi_n^i\rangle\}$.

Spectral decomposition of an observable

Let $\{|\psi_n^i\rangle\}$ be the orthonormal eigenvectors of A with eigenvalues a_n. We define the projectors onto each eigensubspace:

$$P_n = \sum_{i=1}^{g_n} |\psi_n^i\rangle\langle\psi_n^i|.$$

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Then the observable A can be written as:

$$\boxed{A = \sum_n a_n P_n.}$$

This is the **spectral (projector) decomposition** of A.

Example: the projector P_ψ

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Any ket can be decomposed as:

$$|\phi\rangle = P_\psi|\phi\rangle + (1 - P_\psi)|\phi\rangle.$$

Sets of Commuting Observables — Theorem I

Theorem I (Fundamental fact):

If two operators A and B commute, and $|\psi\rangle$ is an eigenvector of A, then $B|\psi\rangle$ is also an eigenvector of A with the same eigenvalue.

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$$A|\psi\rangle = a|\psi\rangle$$

Apply B to both sides:

$$BA|\psi\rangle = aB|\psi\rangle$$

Since $AB = BA$:

$$A(B|\psi\rangle) = a(B|\psi\rangle).$$

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Equivalent Form of Theorem I:
If A and B commute, then every eigensubspace $\mathcal{E}_a$ of A is globally invariant under the action of B.

Theorem II

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$$A|\psi_1\rangle = a_1|\psi_1\rangle, \qquad A|\psi_2\rangle = a_2|\psi_2\rangle.$$

$$\Rightarrow \langle\psi_1|B|\psi_2\rangle = 0 \quad (a_1 \neq a_2).$$

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Subtracting: $(a_1 - a_2)\langle\psi_1|B|\psi_2\rangle = 0$. If $a_1 \neq a_2$:

$$\langle\psi_1|B|\psi_2\rangle = 0.$$

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Let

$$A|u_i^{(n)}\rangle = a_n|u_n^i\rangle, \qquad i = 1, \ldots, g_n.$$

These span the subspace $\mathcal{E}_n$ of dimension g_n.

Structure of the Matrix Representing B

Using Theorem II, the matrix elements of B satisfy

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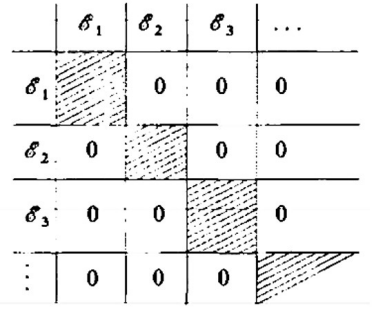
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Case (ii): a_n degenerate ($g_n > 1$):
Block has size $g_n \times g_n$. We must diagonalize B inside $\mathcal{E}_n$ to obtain common eigenvectors.
However, an arbitrary ket of $\mathcal{E}_n$ is an eigenvector of A with the eigenvalue a_n

Constructing Common Eigenvectors

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The set $\{|v_n^i\rangle\}$ forms an orthonormal basis of $\mathcal{E}_n$ consisting of common eigenvectors of A and B.

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$$[A, B] = 0,$$

then one can choose a basis $\{|u^i_{n,p}\rangle\}$ such that:

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Converse of theorem III: if there exists a basis of eigenvectors common to A and B, these two observables commute.
If observable $C = A + B$ with $[A, B] = 0$, the basis $\left|u^i_{n,p}\right\rangle$ common to A and B is also a basis of C, with $\left|u^i_{n,p}\right\rangle$ and eigenvector of C with eigenvalue $a_n + b_p$.

Complete Sets of Commuting Observables (C.S.C.O.)

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This yields a unique orthonormal basis of common eigenvectors.
IMPORTANT: If observable A is non-degenerate, each basis vector can be labelled by its eigenvalue. A would constitute by itself a C.S.C.O.

When Is a Set of Commuting Observables Complete?

However, if A has some degenerate eigenvalues, we would choose another observable B which commutes with A and construct an orthonormal basis common to A and B. If the basis is unique (no degeneracy), then A and B form a C.S.C.O. If degeneracies remain after A and B, we add a third observable C that commutes with both.

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Importance of C.S.C.O.

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They provide the natural basis for solving the Schrödinger equation and classifying quantum states.